

Progress Report: Fish Classification Using Frequency Response Data

Phyllis Sun

1. Introduction

This project aims to classify fish species using sonar-based frequency response data. The dataset consists of measurements taken from two fish species: Lake Trout (LT) and Smallmouth Bass (SMB). Classifying these two species is important for fisheries management, conservation efforts, and ecological monitoring in Ontario's freshwater ecosystems, including the Great Lakes and inland lakes near Toronto.

Lake Trout populations have historically declined due to overfishing, invasive species, and climate change, while Smallmouth Bass are expanding their range, potentially disrupting native Lake Trout habitats. Therefore, understanding their distributions is critical for monitoring ecosystem shifts and guiding conservation strategies.

2. Data Section

The dataset contains 10,107 observations and 276 variables after cleaning, where repeated frequency tests were conducted per fish. By placing a fish underwater, the frequency data was collected using a sonar system, where sound waves were transmitted from a boat, bounced off a fish, and received back.

The key features include:

- **Fish ID:** Unique identifier for each fish. (fish_Num)
- **Species:** The categorical variable representing Lake Trout or Smallmouth Bass (species)
- **Frequency Variables (F45-F260):** Numeric values representing the frequency response measurements. (F45, F45.5,...,F260)
- **Biological Features:** Total length, fork length, weight, girth, dorso-lateral height, sex, air bladder total length, air bladder width, and air bladder weight.

Data Preprocessing

The original dataset has 14,575 observations and 484 variables. It includes three fish species (Lake Trout, Lake Whitefish, and Smallmouth Bass), time-related data (e.g., dateTimeSample, dateProcessed, Ping_time), and spatial data (e.g., Target_depth_mean, Target_range). However, since I am only using Lake Trout and Smallmouth Bass for classification, I excluded Lake Whitefish. Additionally, because the fish were manually placed underwater, the spatial data appears artificial. Therefore, after feature selection, I created the following two datasets:

- **fish_data:** Includes biological features alongside frequency response data to analyze their relationship.
- **freq_only:** Contains only fish labels and frequency responses to focus on species-specific patterns.

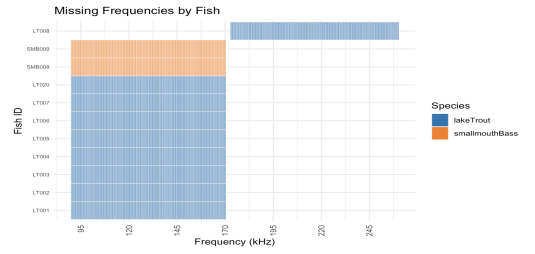


Figure 1. Missing frequency per fish Heatmap

After feature selection, I counted the missing values for each fish and identified which frequency ranges were missing using a heatmap (Fig. 1). The heatmap clearly shows that most fish are missing data in the 90–170 kHz frequency range, while only the fish with ID LT008 has data missing in the 173–260 kHz range, so I did the following action for handling the missing values:

- The 90–170 kHz frequency range was removed due to excessive noise and inconsistencies.
- The fish with ID LT008 was removed since it had over 50% missing values.

I also checked all variable types and found that the Frequency Response and Sex variables were stored as characters. I converted them to numeric for easier use in the analysis.

In my Exploratory Data Analysis (EDA) step:

1. I first examined the Mean Frequency Response Strength Patterns per Fish group by species and observed that Lake Trout exhibited stronger responses than Smallmouth Bass across most frequency ranges, with clearer separation in the 45–90 kHz range.
2. For the T-Test Analysis, after reviewing my teacher’s comment on my EDA report regarding Bonferroni correction and confirming that the frequencies are correlated, I decided not to use the T-test analysis.
3. I also checked for outliers in the Frequency Response variable but found that the difference in mean strength per frequency per fish did not vary significantly. Figure 2 shows that most differences are close to 0, with only a few ranging between 2–3. Therefore, I decided not to remove these outliers

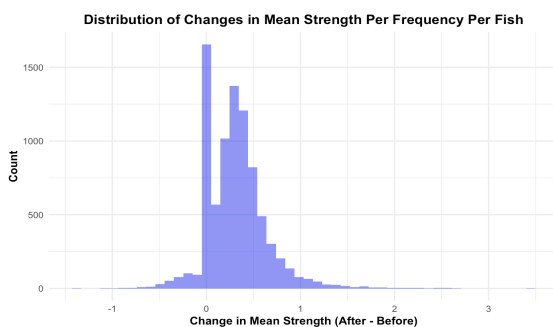


Figure 2. Histogram for distribution of differences in mean strength

- Then, I used Principal Component Analysis (PCA). From Figures 3 and 4, we observe that the most important frequencies fall within the 63–77 kHz range, contributing significantly to PC1, which explains 48.6% of the total variance. These frequencies likely capture broad variations in fish frequency responses, but their role in distinguishing species remains uncertain. However, I will keep the PCA frequency range in the modeling step for exploratory purposes.

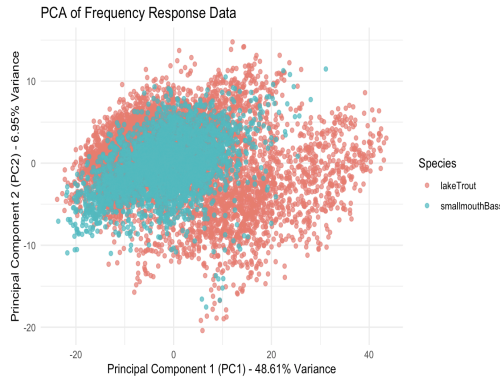


Figure 3. PCA scatter plot

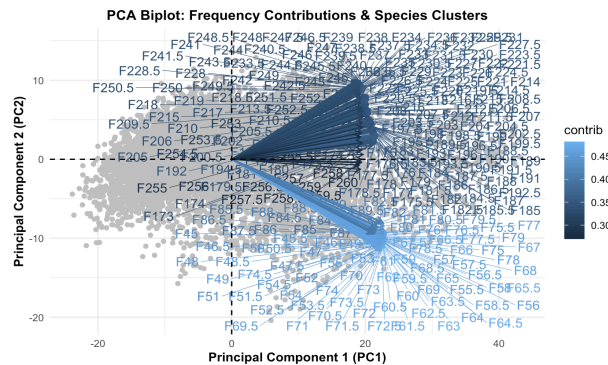


Figure 4. PCA Biplot

- I also conducted a Correlation Analysis. Figure 5 shows that adjacent frequencies exhibit strong correlations, highlighting redundancy in some signals. Figure 6 illustrates that correlations between biological features and frequency responses vary by species, with Lake Trout responses being more influenced by air bladder width and Smallmouth Bass by body size. Therefore, I can use key biological features that vary by species in the modeling process.

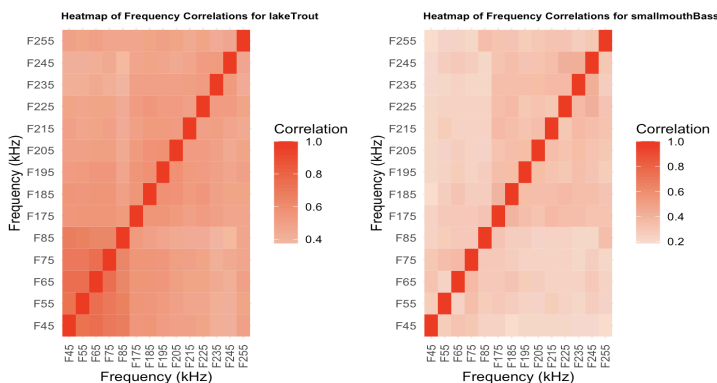


Figure 5. Correlation between frequency heatmap

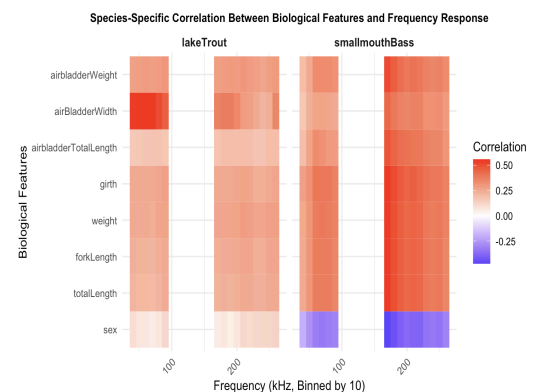


Figure 6. Biological Feature VS. Frequency Heatmap

- The Sex-Based Comparison step is not central to the project objective, as our main goal is to use frequency response data. Therefore, I decided not to explore this approach further.

3. Method Section

Data Preparation:

To evaluate the impact of different features on classification performance, I constructed six variations of the dataset. Each dataset includes different combinations of biological features and frequency

response data:

1. All Biological Features + All Frequency Response Data

- Includes all available biological measurements (e.g., length, weight, girth, air bladder characteristics) along with frequency response features.

2. Species-Specific Biological Features + All Frequency Response Data

- Uses different biological features for each species while keeping all frequency response data. For example, some species may have additional measurements like girth and sex, while others may not.

3. PCA-Selected Frequency Range + All Biological Features

- Applies **Principal Component Analysis (PCA)** to identify key frequency components, combined with all biological features.

4. PCA-Selected Frequency Range + Species-Specific Biological Features

- Uses PCA-determined frequency features while selecting different biological features for each species.

5. Only Frequency Response Data

- Excludes biological features and relies solely on frequency response data for classification.

6. Only PCA-Selected Frequency Range

- Excludes biological features and uses only the PCA-selected subset of frequency response data.

Next, to address the research question of classifying fish species based on sonar frequency response data, I employ the SuperLearner ensemble method as our primary modeling approach. Given the relatively small sample size ($\leq 10,000$ observations), SuperLearner is an appropriate choice because it allows us to run multiple models simultaneously, utilizing their strengths to improve accuracy while reducing overfitting. Instead of selecting a single model a priori, SuperLearner uses 5-fold cross-validation to determine the optimal weighted combination of base models, ensuring better generalization to unseen data.

The characteristics of the dataset drive the choice of base learners. I select the following models for inclusion in the ensemble:

- SL.glmnet (Elastic Net Regression): Suitable for high-dimensional data with correlated features, such as frequency response values at different kHz levels.
- SL.randomForest (Random Forest): Captures non-linear interactions between biological and frequency-based features while handling missing data effectively.
- SL.xgboost (Extreme Gradient Boosting): A powerful tree-based model that optimizes feature selection and provides robustness against noise.

- SL.nnet (Neural Networks): Learns complex hierarchical patterns from the data, useful when subtle variations in sonar responses differentiate species.
- SL.ranger (Fast Random Forest Implementation): Optimized for efficient handling of large feature spaces, which is crucial given the number of frequency response variables.

I use a binomial family model (family = binomial()) because the classification task involves distinguishing two species (Lake Trout vs. Smallmouth Bass). The choice of NNLS (Non-Negative Least Squares) as the meta-learner ensures that each base learner's contribution remains non-negative, preventing overfitting by assigning extremely negative weights. This guarantees that the ensemble prediction is a weighted average of individual models, enhancing stability and interpretability.

To evaluate the performance of the trained model, I used multiple classification metrics, including accuracy, balanced accuracy, and AUC-ROC score. The model predictions were compared against the true labels from the test dataset, and a confusion matrix was generated to analyze misclassifications.

Since the dataset may have class imbalances, balanced accuracy was calculated as the average of sensitivity (true positive rate) and specificity (true negative rate), ensuring a fair evaluation across both classes. The AUC-ROC score was also computed to assess the model’s ability to distinguish between fish species. A higher AUC value indicates stronger classification performance.

4. Results

From this point forward, I trained the model using only frequency response data and obtained the following results. The XGBoost model had the largest weight (0.489) and a relatively low cross-validated (CV) risk, indicating its strong performance in the SuperLearner framework.

Given these results, I proceeded to train XGBoost independently and evaluated its performance, as shown in List 1. The model achieved a high balanced accuracy of 77.7% which indicates that the model performs well in distinguishing between the two fish species while accounting for potential class imbalances. The high AUC-ROC score (89.2%) suggests strong discriminatory power, meaning the model effectively differentiates between species based on their frequency response patterns. These results demonstrate that frequency response data alone provides a solid foundation for classification. However, further improvements could be explored through feature selection, hyperparameter tuning, or incorporating biological features to enhance classification accuracy.

	Algorithm <chr>	Weight <dbl>	CV_Risk <dbl>
SL.glmnet_All	SL.glmnet_All	0.000000	0.1942185
SL.randomForest_All	SL.randomForest_All	0.293400	0.1340113
SL.xgboost_All	SL.xgboost_All	0.489331	0.1302807
SL.nnet_All	SL.nnet_All	0.000000	0.2461085
SL.ranger_All	SL.ranger_All	0.217269	0.1341505

Balanced Accuracy: 0.777
 Setting levels: control = 0, case = 1
 Setting direction: controls < cases
 AUC-ROC Score: 0.892

Table 1. Only Frequency Data Result from Superlearner

List 1. XGBoot model evaluation result

5. Next Step

Next, I will test the remaining dataset variations in SuperLearner to identify the best feature combination for fish classification. After evaluating model performance using balanced accuracy, and AUC-ROC, I will select the top dataset-model pair. Finally, I will fine-tune the best model through hyperparameter optimization to maximize classification accuracy.